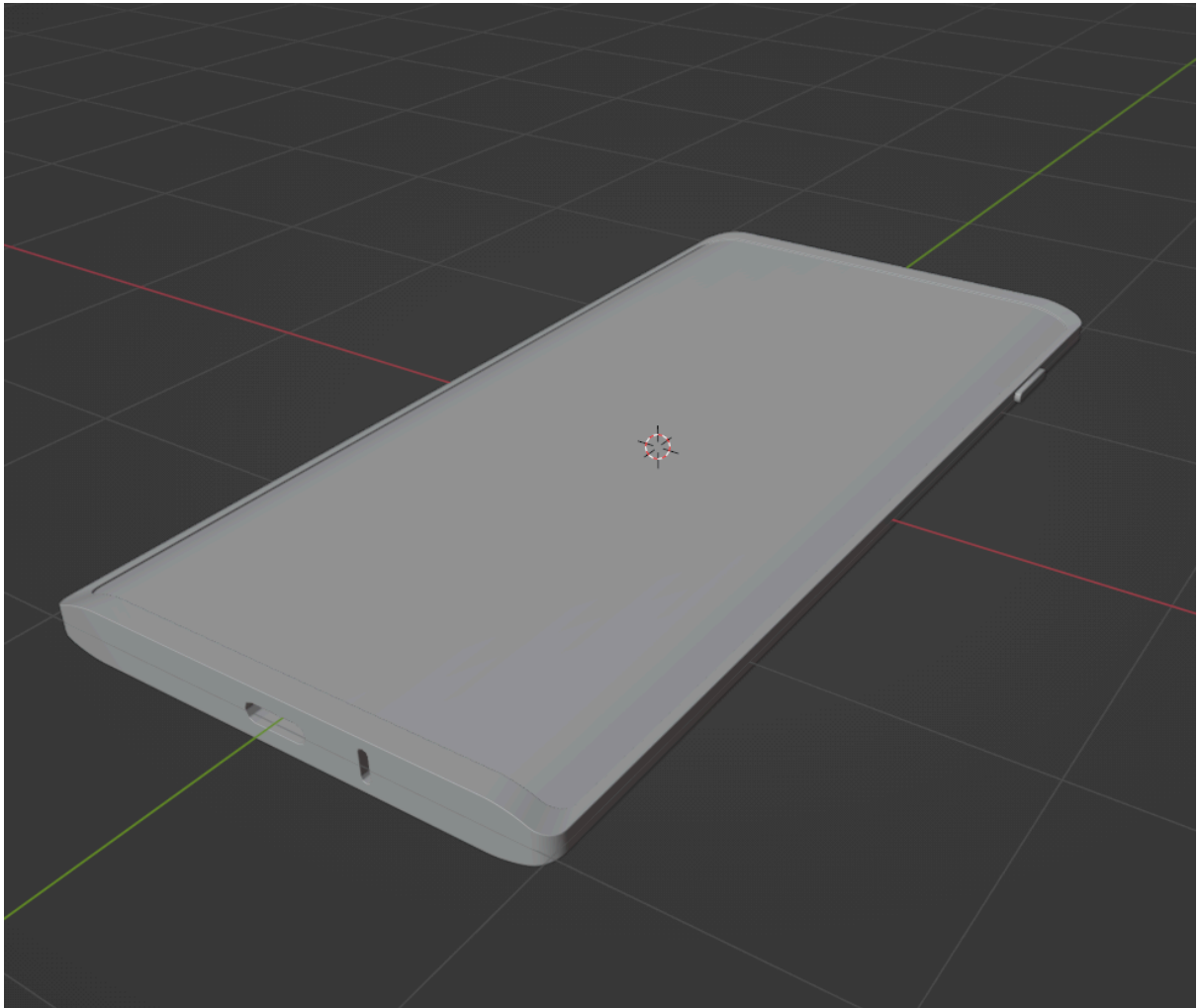
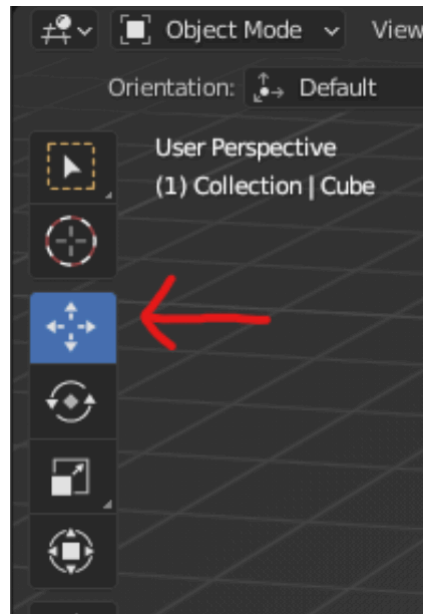


Difficult tutorial 1

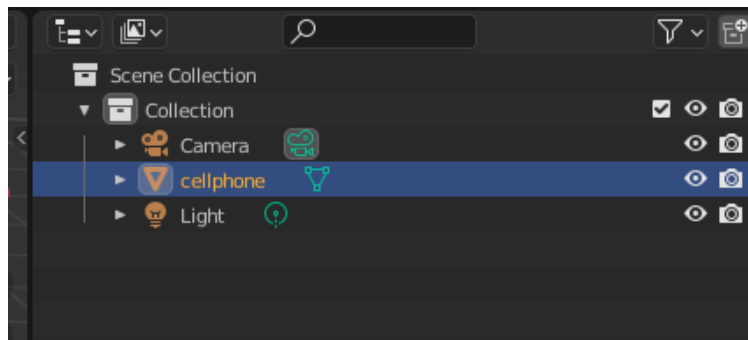


In this exercise, we are going to create a detailed modern cellphone.

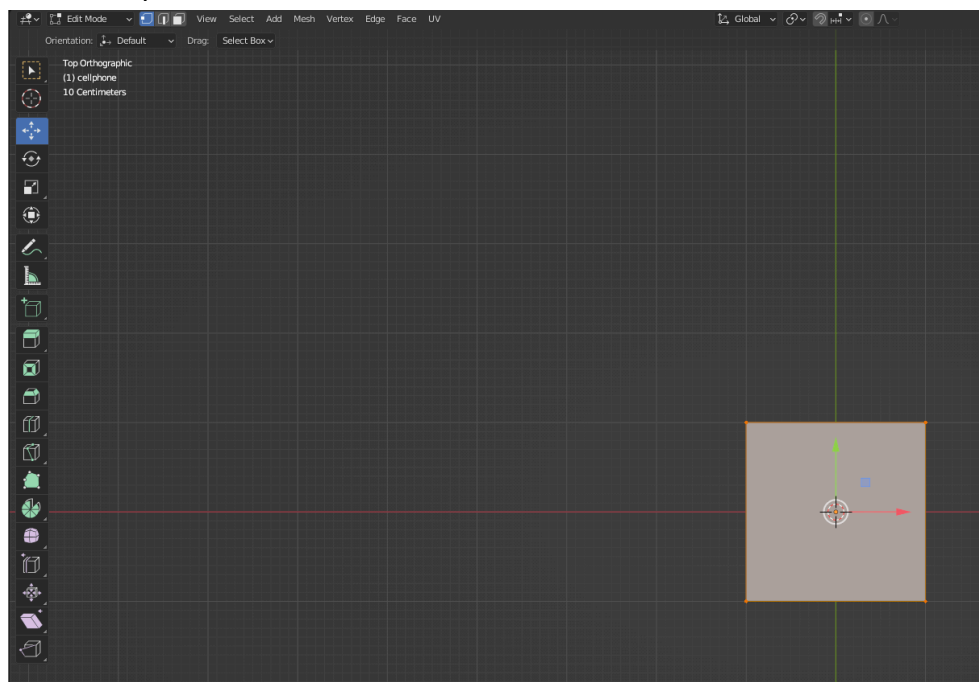
To start, set the viewport mode to show arrows of the axis to aid in the modeling process:



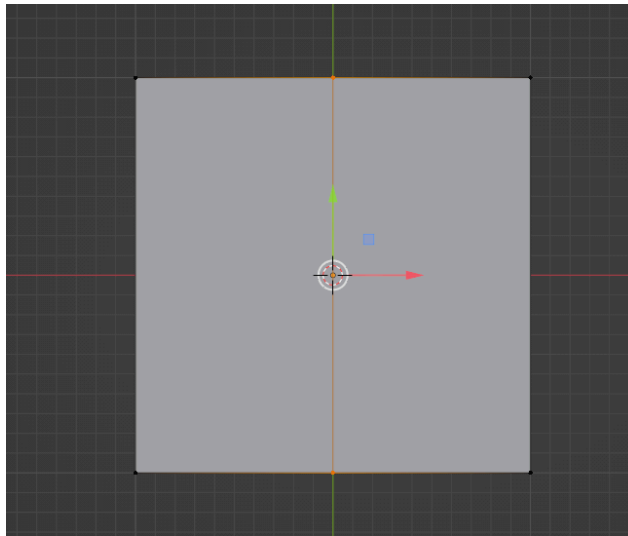
Rename the default cube to “cellphone” (exactly like this, case sensitive):



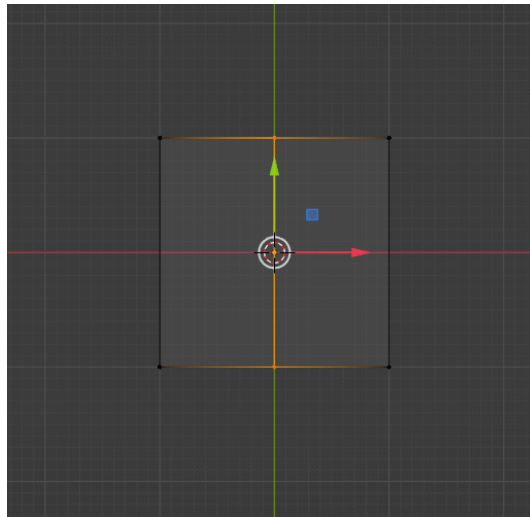
Go to the top view (NUMPAD 7) and enter edit mode (TAB).
Again, set the viewport mode to show arrows of the axis:



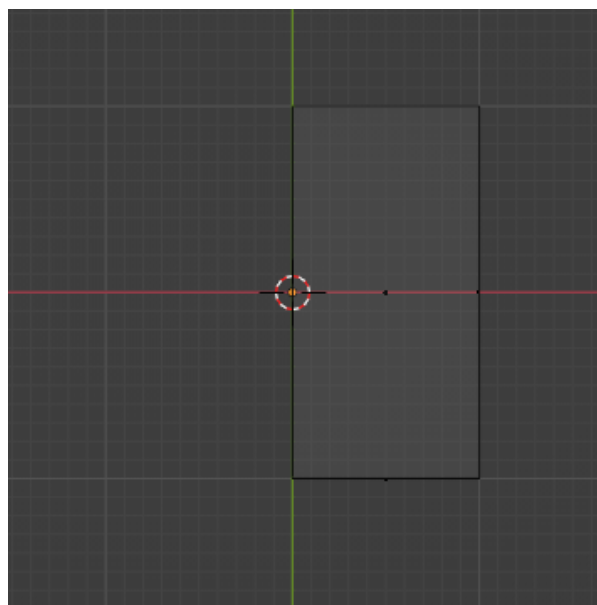
Now, apply a loop cut (CTRL+R) in the middle of the cube:



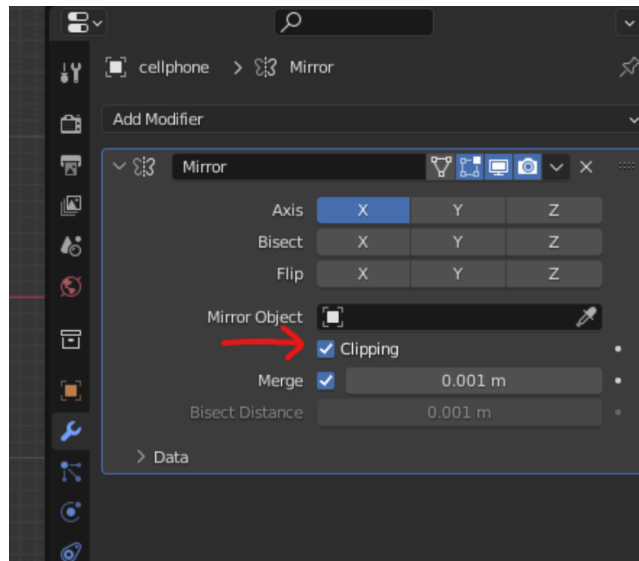
Change to wireframe view (press Z and then select Wireframe).



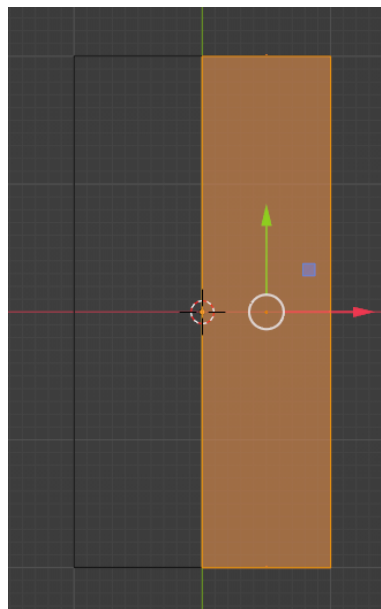
Change to face mode (press number 3) and select the left side of the cube and delete faces:



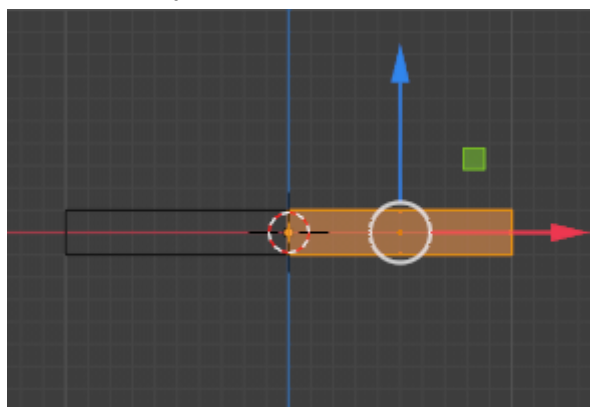
Next, add a mirror modifier (check if Axis = X) and enable clipping:



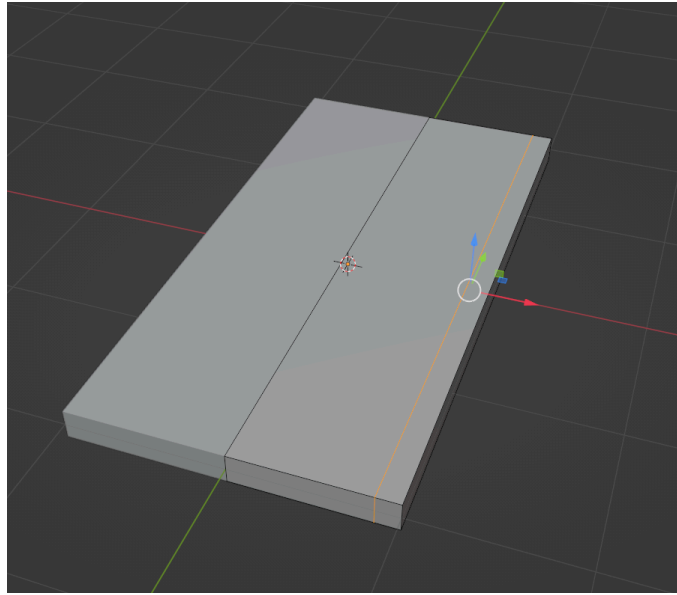
Select all (press A) and scale on the Y axis (S+Y) by 2:



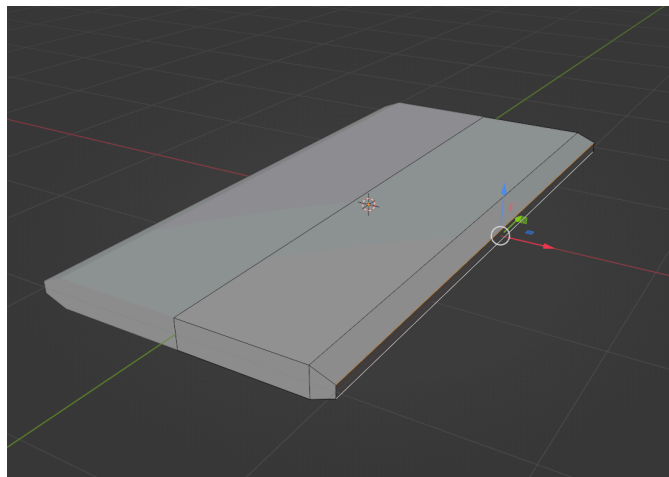
Go to front view and scale Z (S+Z) by 0.1:



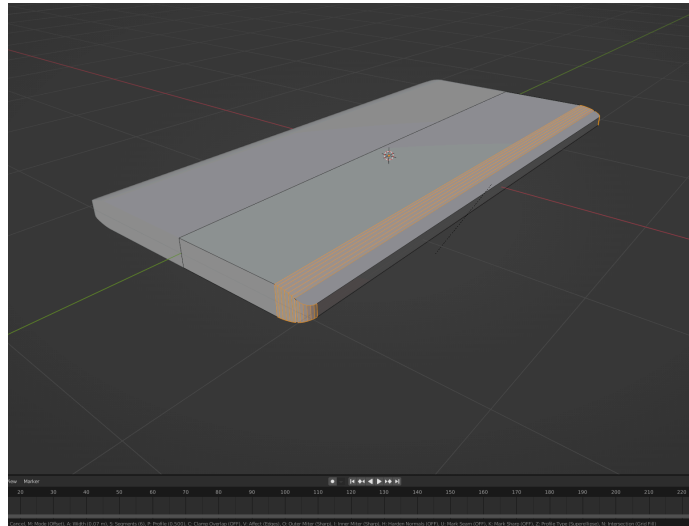
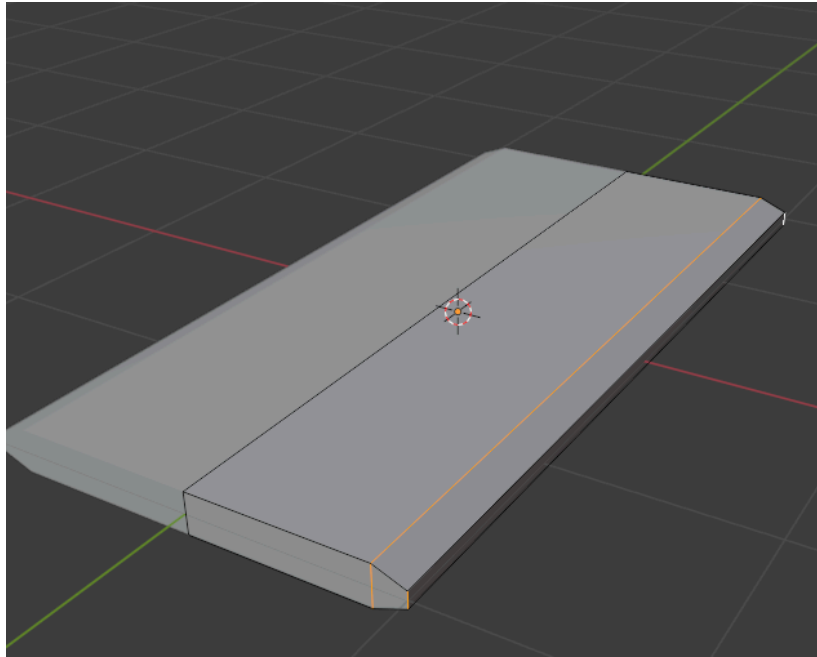
Change back to solid shade (press Z and select solid) and add another loop cut (CTRL+R) in y direction typing 0.7 so it is moved towards the outside (if it moves to the center, type -0.7):



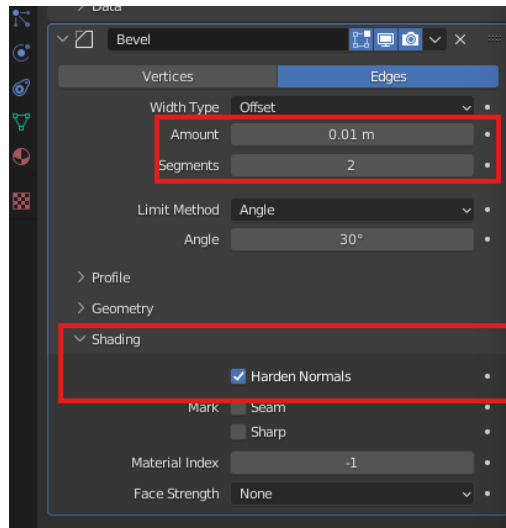
Select the outer horizontal edges and scale in Z (s+Z) direction by 0.4:



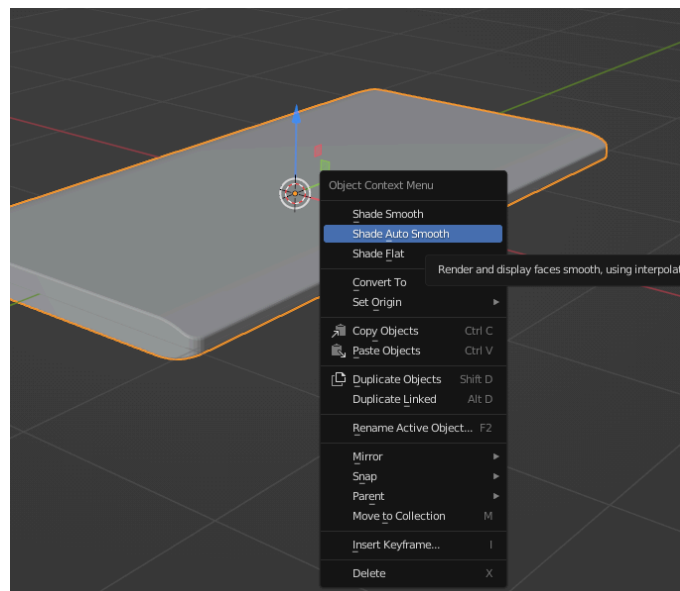
Select the loop created (LMB while holding ALT on the edge) and the outer vertical edges and bevel (CTRL+B) by 0.07 and increase the number of segments to 6 using the mouse wheel



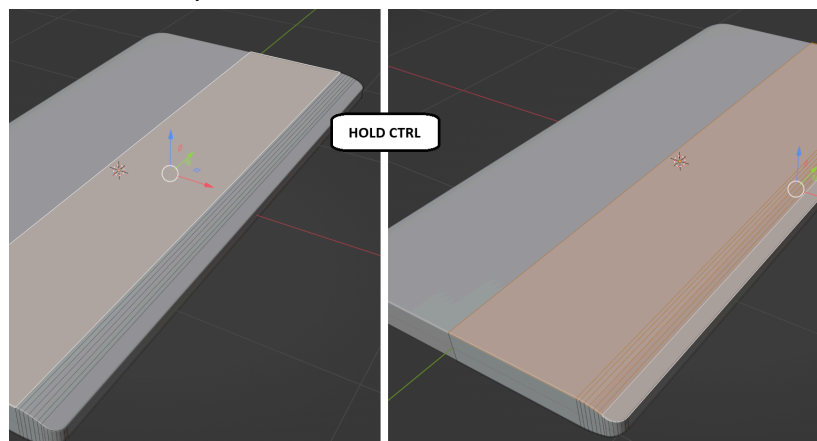
Add a Bevel modifier and set the Amount to 0.01 and Segments to 2.
Under shading, select also “Harden Normals”:



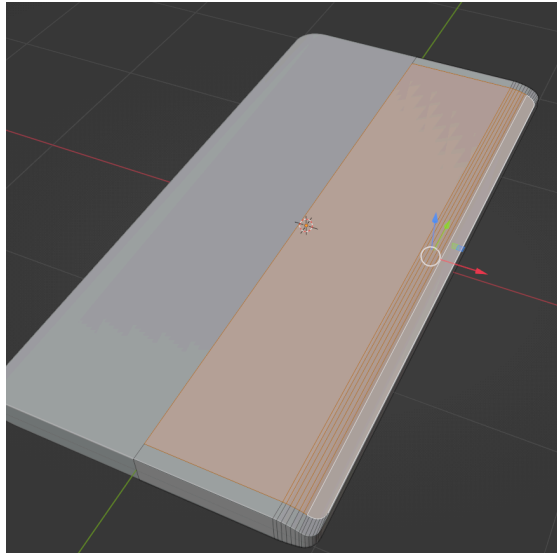
Enter Object mode (TAB), right click and select “Shade Auto Smooth”:



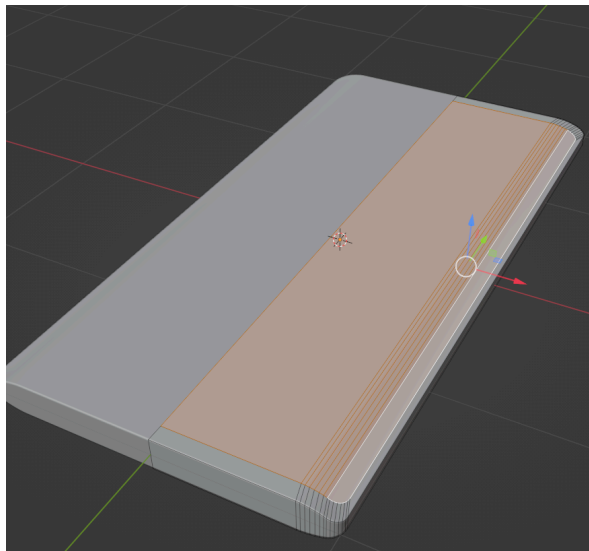
Go back to edit mode, change to Face selection mode and select all the top faces. You can do this by selecting the leftmost face, holding CTRL and selecting the rightmost face, so it will select all the faces in the path:



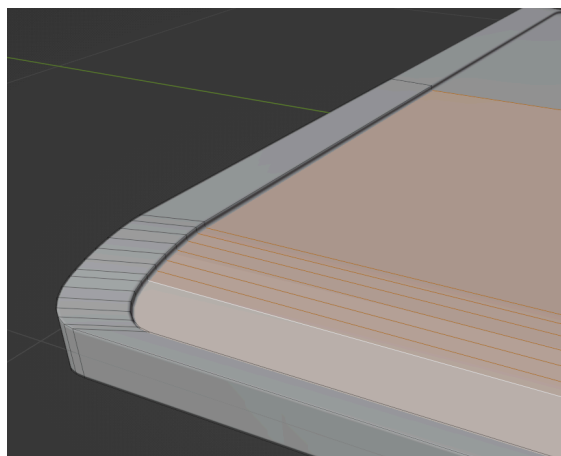
Extrude and confirm (E+ENTER). Then, scale the selection (S) by 0.95:



Move the selection in the X (G+X) axis by -0.03:



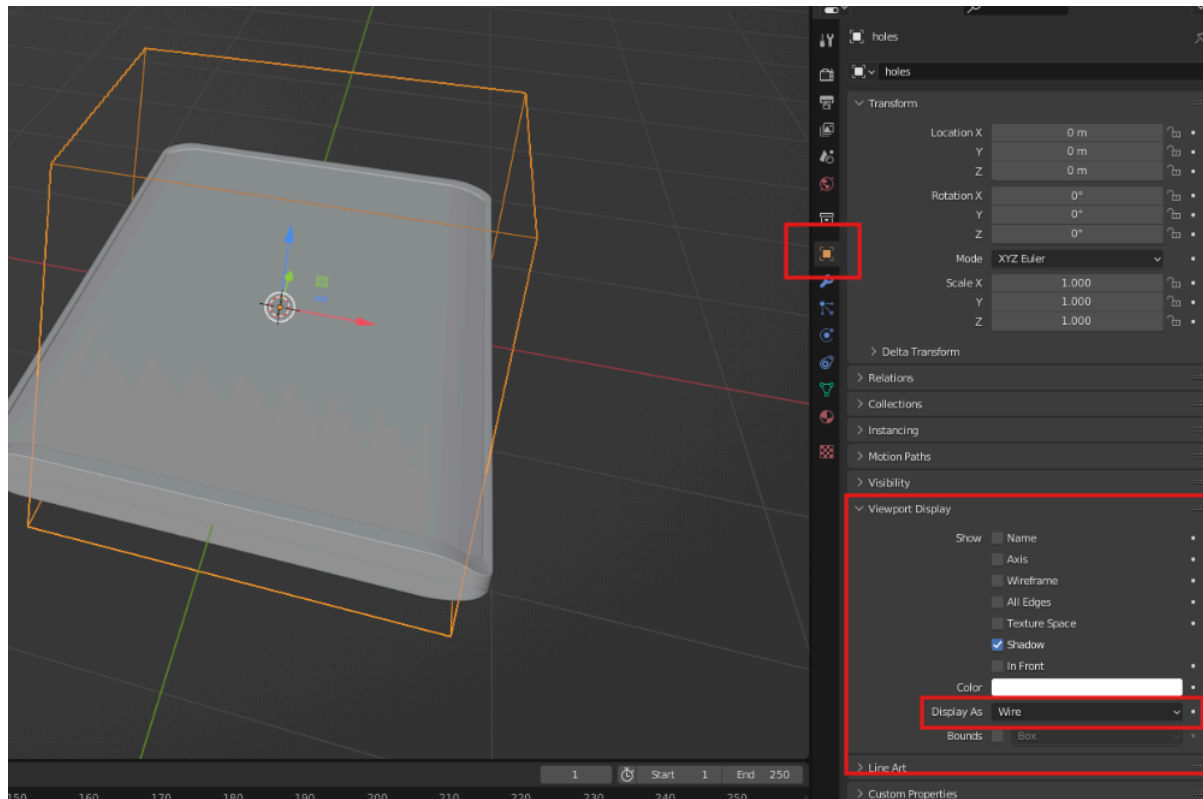
Extrude again (E+ENTER) and move in the z (G+Z) axis by -0.01 to create the screen depth:



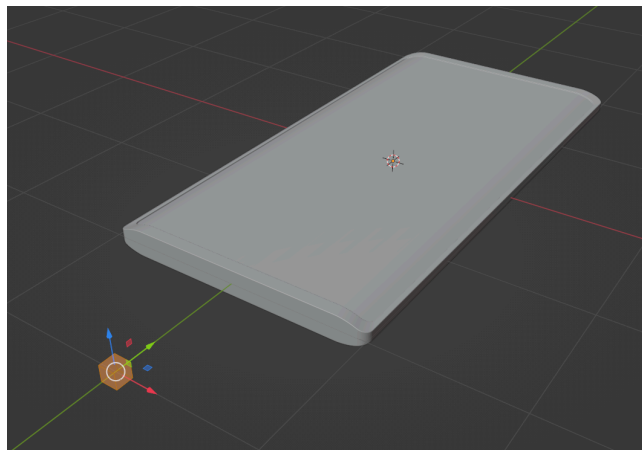
Now, we are going to start detailing the cellphone. This will be done using boolean operators.

Go back to object mode (TAB).

Create a new cube mesh (Make sure that the 3D cursor is at the center of the scene) and name it “holes” (SHIFT+A → Mesh → Cube). Then go to the Object Properties panel, go to Viewport Display and select “Wire” under “Display as” property

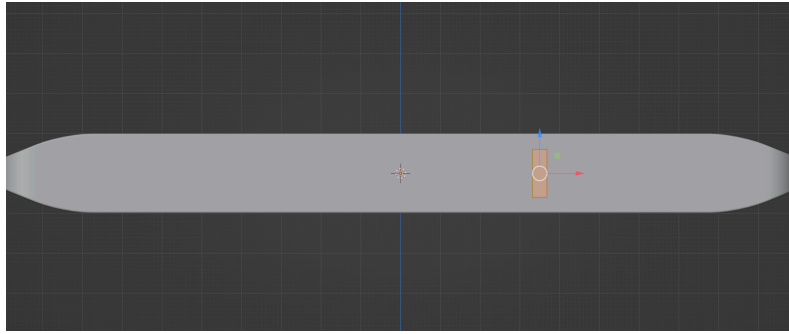


Enter edit mode (TAB) in the new object created (holes) and move the cube mesh in the Y axis (G+Y) by -3. Then, Scale it down (S) by 0.06

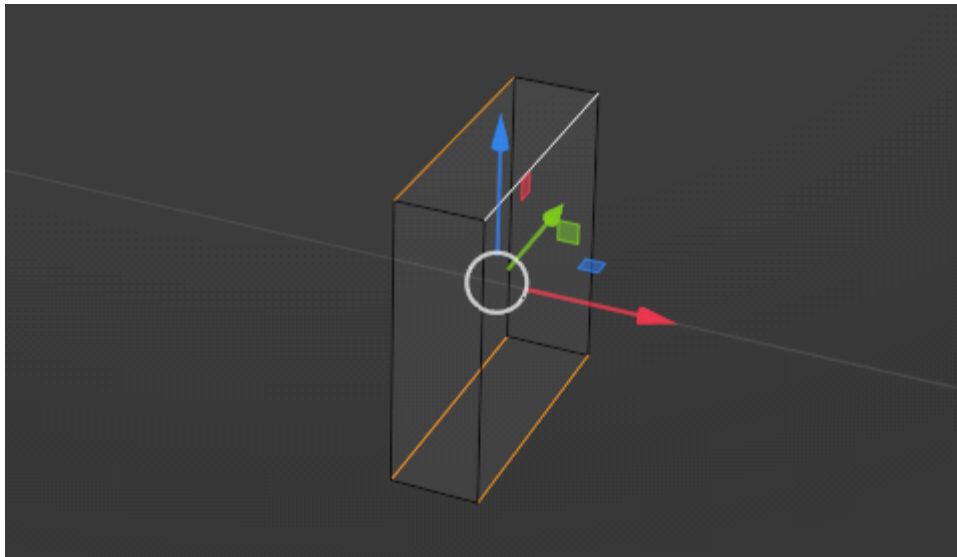


Go to the Front view (NUMPAD 1) and scale in the X (S+X) direction by 0.3.

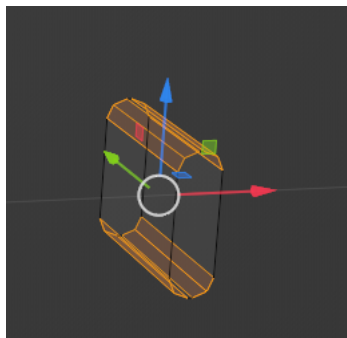
Next, move it in the X (G+X) direction by 0.35



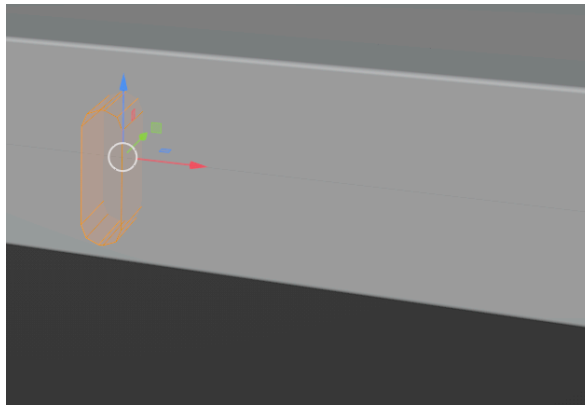
Change to edge selection (press number 2) and select all the horizontal edges of the mesh. You can do this by going back to perspective view, holding CTRL + ALT and selecting one of the horizontal edges. Blender will automatically select all parallel edges as a loop:



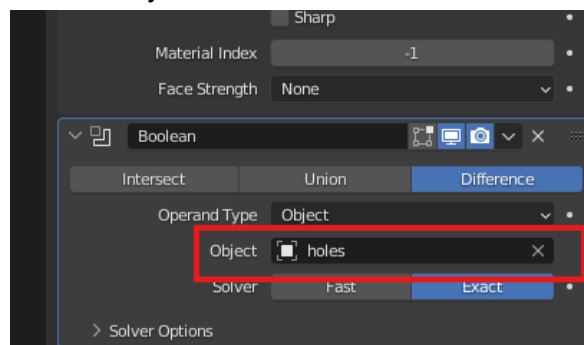
Bevel the edges (CTRL+B) by 0.016 and using 2 segments:



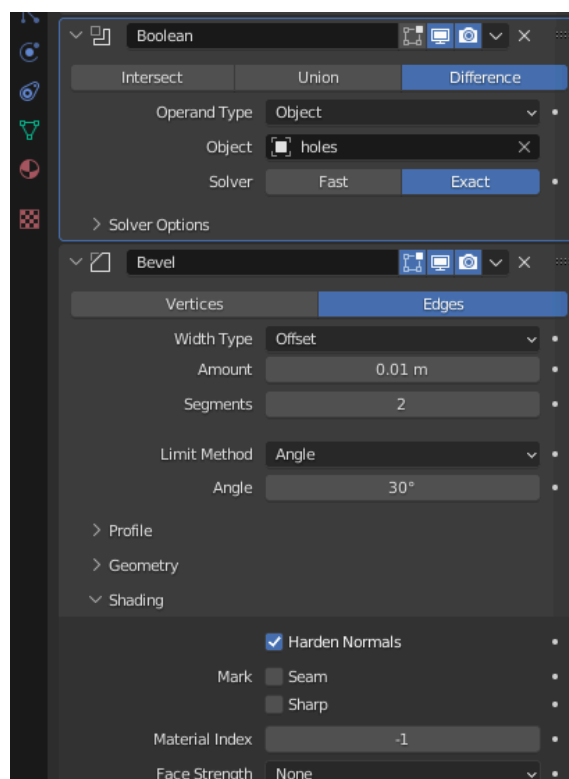
Now select all (A) and move in the Y axis (G+Y) by 1 so it is half inside the phone.

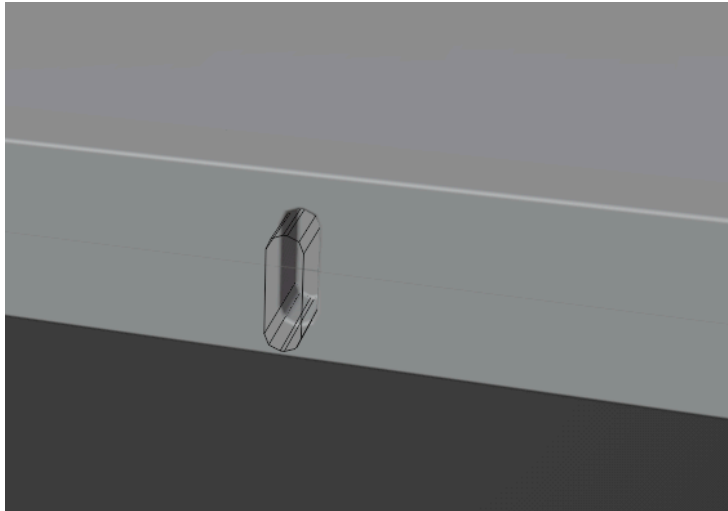


Now go to object mode, select the cell phone and add a boolean modifier. Select the Object as the “holes” object



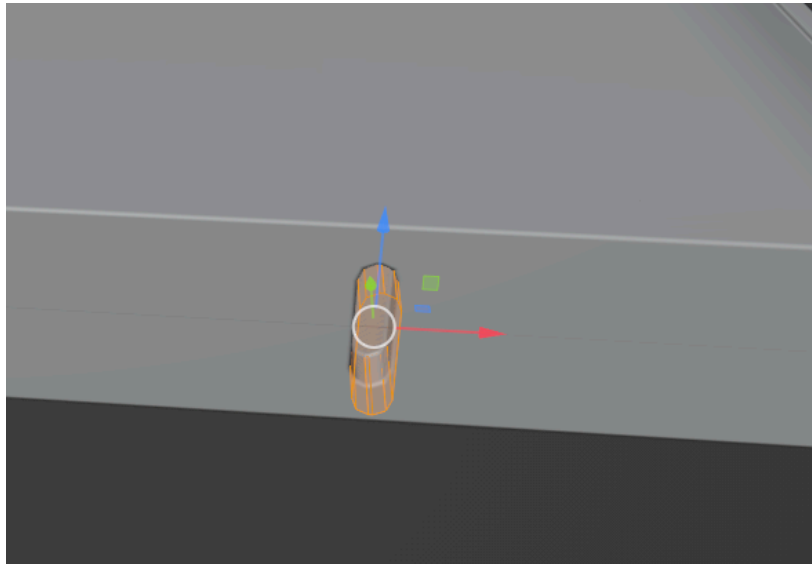
In order to make it more realistic, move the Boolean modifier up to be applied before the Bevel modifier:



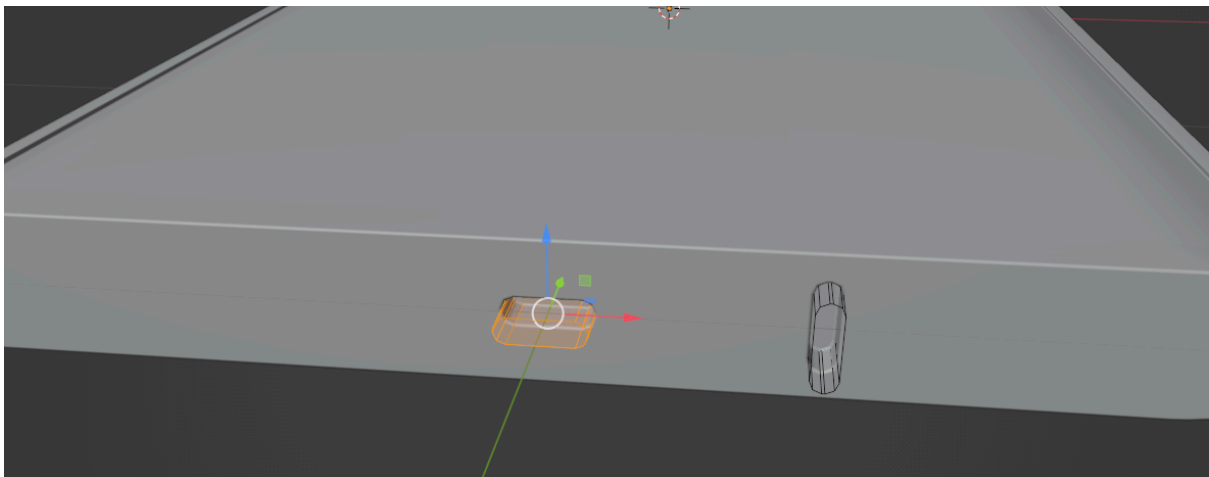


Now select the holes object again, enter edit mode (TAB) and select the rounded mesh created.

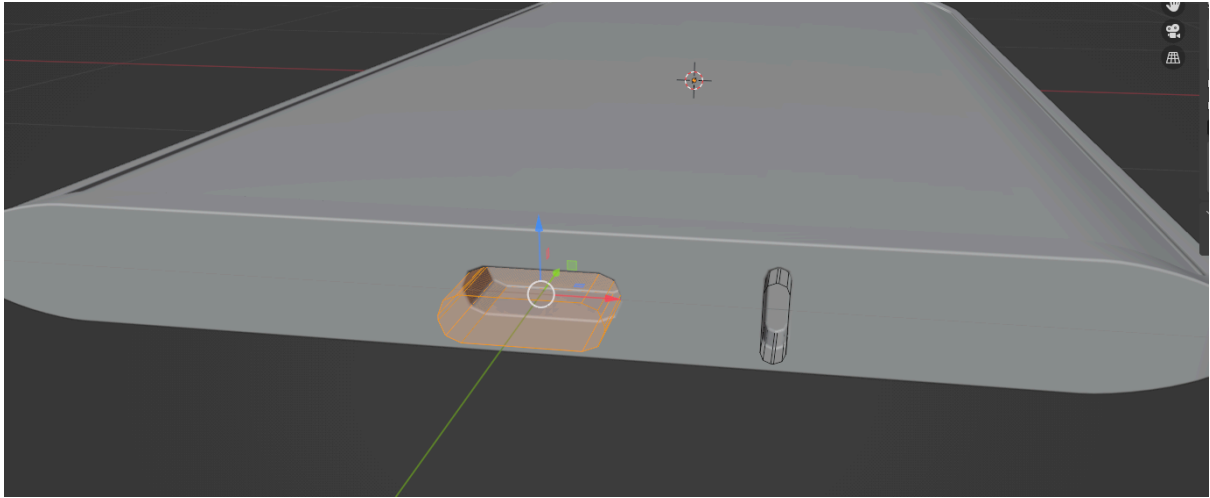
You can do this by hovering the mouse on top of the mesh and typing “L” to select all linked mesh:



Duplicate it (SHIFT+D), move on the X axis (G+X) by -0.35 to center it. Now rotate it in the Y axis (R+Y) by 90 degrees.



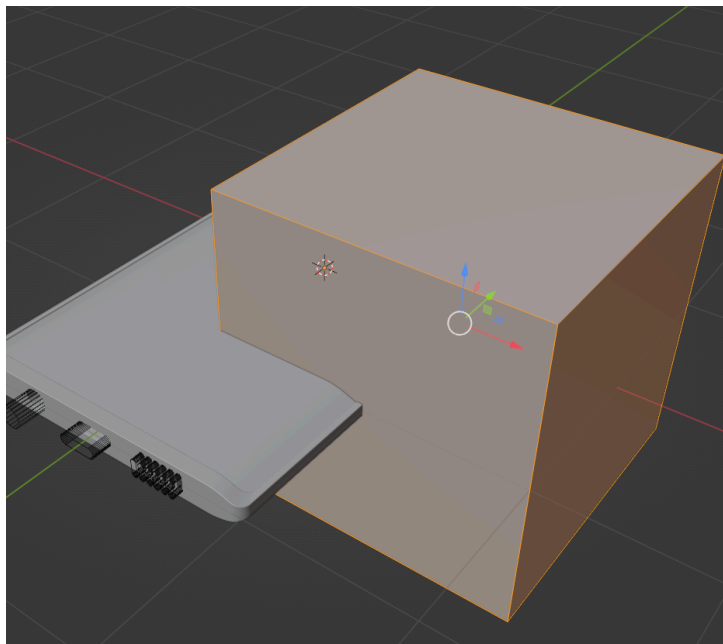
As you can imagine, this will be the USB-C input charger of the phone. But it is too small. So scale it up in all directions (S) by 2:



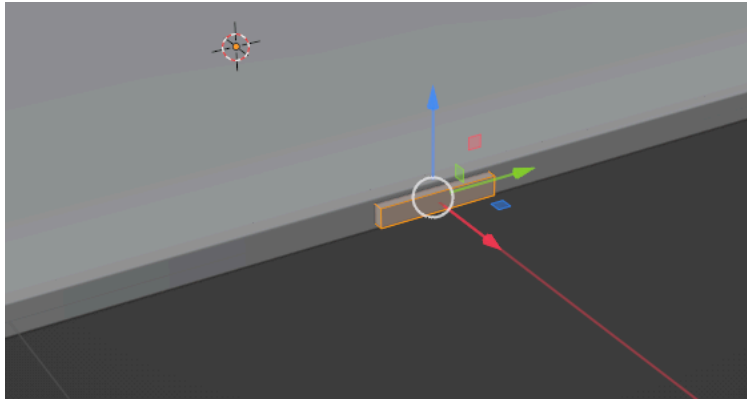
Now we are going to add a button to the side of the phone.

Go back to object mode and create a new Cube object mesh (SHIFT+A → Mesh → Cube), rename it as “button” and enter edit mode.

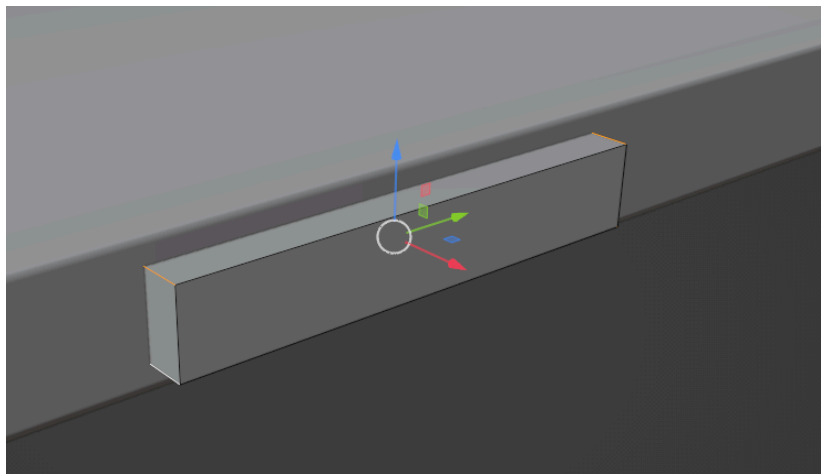
Move the cube in the X (G+X) axis by 1:



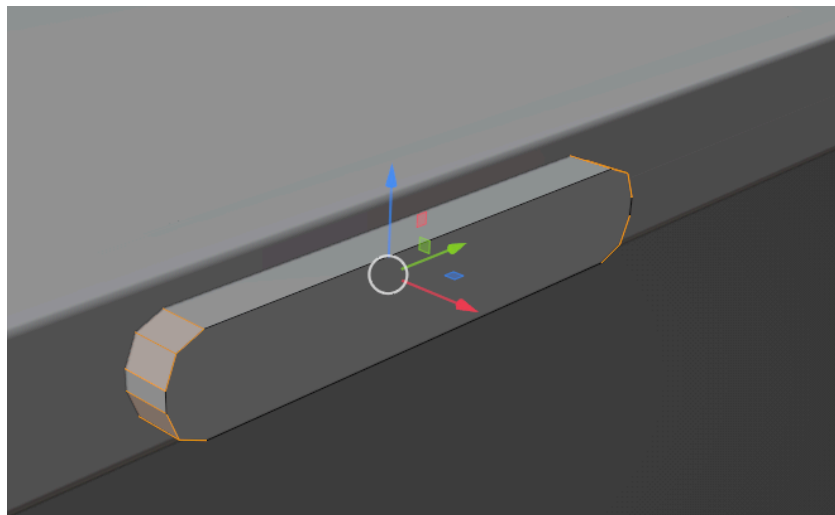
Scale the cube (S) by 0.025. Then, scale it in the Y axis (S+Y) by 6:



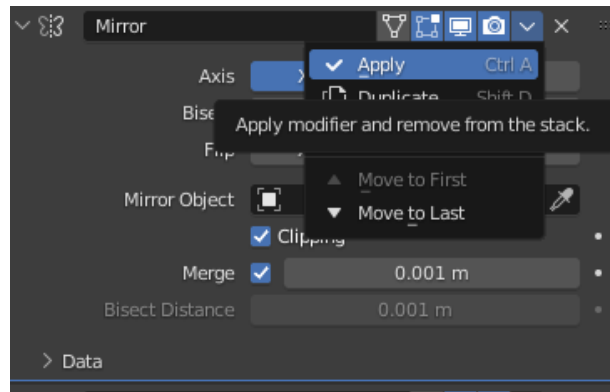
Then move it in the Y axis (G+Y) by 1.3 to move it to the top portion of the phone.
Select all horizontal edges:



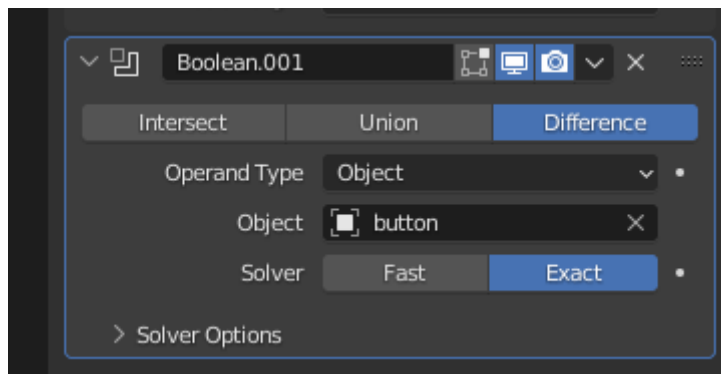
and bevel them (CTRL+B) by 0.02 and using again 2 segments in order to make it curved:



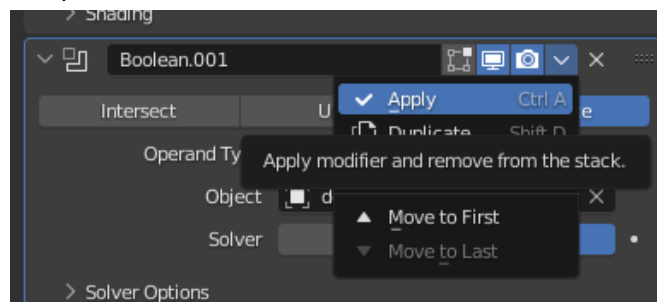
Now go to the object mode (TAB), select the phone and apply the mirror modifier:



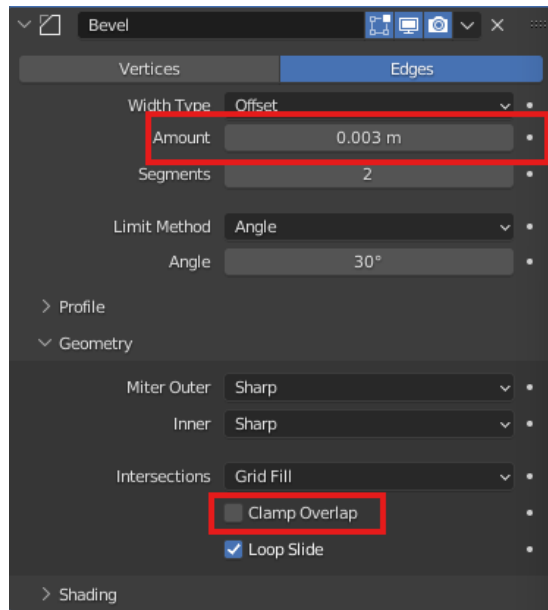
Then, add another boolean modifier and select the “button” object created.



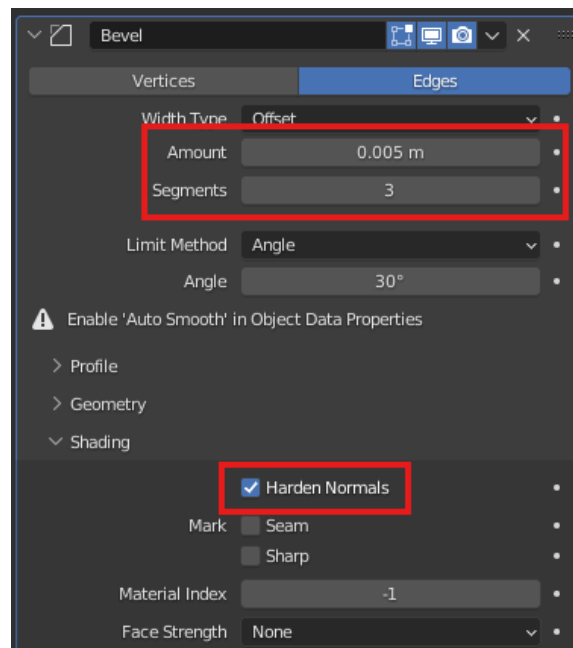
Now apply the boolean operator for the button:



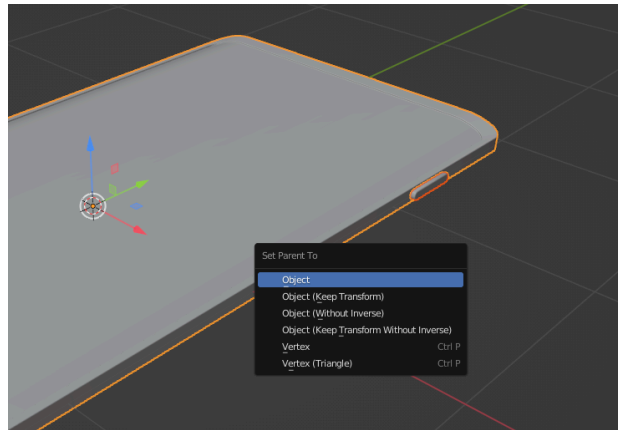
In the bevel modifier of the cellphone, set the “Amount” value to 0.003 and disable “Clamp Overlap” under “Geometry” options:



Now select the button again and add a bevel modifier.
Set the Amount to 0.005, the number of segments to 3 and enable “Harden Normals” under the “Shading” tab.



Now, right click the viewport and select “Shade auto smooth” with the buttons selected.
With the buttons selected, select also the phone while holding SHIFT and parent (Set Parent to Object) the buttons to the phone:



Now, select the phone and apply also the boolean modifier for the holes and delete the holes object:

